SET 3: THE XANADU HIGHLANDS DEEP-ICE GEOTHERMAL TAP & CRYO-AMMONIA PLANT

DOCUMENT 3A: SECURED CREDIT AGREEMENT

SECURED TERM CREDIT AGREEMENT

Dated as of the Agreement Effective Date (as defined herein)

among

XANADU CRYO-CHEMICALS LP,
as Borrower

and

MARTIAN COLONY INVESTMENT SYNDICATE (MCIS),
as Lead Arranger, Administrative Agent, and Collateral Agent

and

THE LENDERS PARTY HERETO FROM TIME TO TIME

This SECURED TERM CREDIT AGREEMENT (this "Agreement") is entered into on the Agreement Effective Date by and among the Borrower, the Lenders, and the Administrative Agent. In consideration of the mutual covenants and agreements herein contained, the parties hereto agree as follows:

ARTICLE I: DEFINITIONS AND INTERPRETATION

Section 1.01. Defined Terms. As used in this Agreement, the following terms have the meanings specified below:

- **"Agreement Effective Date"** means the date on which this Agreement has been fully executed and delivered by all parties hereto, as verified by the date inscribed next to the signatures of the authorized officers in the execution block below (such date, the **"Signature Date"**).

- **"Ammonia Synthesis Target"** means the minimum operational production metric of 25,000 metric tons of pure liquid ammonia (NH_3) per Titan sidereal day, as calculated under the provisions of Section 4.01(2) herein.

- **"Cryo-Geothermal Tap Integrity Index"** or **"CGTII"** means the structural, thermodynamic, and pressure stability rating of the deep-bore geothermal tap at Xanadu Highlands, as evaluated by the

Titan Technical Safety Directorate (TTSD).

- **"Environmental Reclamation Reserve"** means the cash escrow reserve fund mandated to be maintained under Section 4.01(3) for localized clathrate stabilization.

- **"Sol System Interstellar Accounting Standards"** or **"SSIAS"** means the unified framework of financial accounting principles approved by the Interstellar Financial Board.

- **"TIDOR"** means the Titan Inter-Dome Offered Rate, which is the floating interbank benchmark rate calculated for Titan-denominated credit lines.

Section 1.02. Rules of Interpretation. Capitalized terms defined in the singular shall have the corresponding meanings in the plural, and vice versa. Any references to specific sections, tables, or documents relate directly to this Agreement unless otherwise specified.

ARTICLE II: THE SECURED CREDIT LINES

Section 2.01. The Facility. Subject to the terms and conditions of this Agreement, each Lender agrees to make a single advance term loan (each a "Facility Loan") to the Borrower on the **Closing Date** (which is identical to the **Signature Date**) in an aggregate principal amount of **3,800,000,000 Sol Credits (SC)**.

Section 2.02. Interest Rate Adjustments.

1. The Facility Loans shall accrue interest daily at a floating rate per annum equal to **TIDOR plus 5.25%**.
2. Should the **Cryo-Geothermal Tap Integrity Index (CGTII)** fall below 82.5% for any consecutive 5-day cycle, the floating margin shall automatically increase by **2.00% per annum** (the "Thermal Risk Penalty Premium") until the CGTII is restored to 90.0% or above, as verified by the TTSD.

Section 2.03. Repayment and Amortization. Repayment of the principal shall be executed in equal quarterly installments on the final operational day of each Martian fiscal quarter, starting on the first such date following the **Agreement Effective Date**, in accordance with the amortization schedules appended hereto.

ARTICLE III: REPRESENTATIONS AND WARRANTIES

The Borrower represents and warrants to the Lenders that:

Section 3.01. Subsurface Mineral Rights. The Borrower holds exclusive subsurface licenses to extract nitrogen gas ballast, water-ice substrate, and ammonia clathrates within the designated coordinate grid (Sector 9-Xanadu) under the authority of the Titan Regulatory Council (TRC).

Section 3.02. Thermal Runaway Mitigation. The cryo-geothermal borehole casing is engineered to withstand pressures up to 900 MPa and thermal gradient surges up to 350 Kelvin, minimizing any localized clathrate destabilization risks.

ARTICLE IV: COVENANTS

Section 4.01. Affirmative Covenants. So long as any part of the Facility Loan remains outstanding, the Borrower shall:

1. **Thermal Tap Maintenance:** Undertake continuous acoustic-imaging diagnostics of the primary geothermal tap borehole to ensure the CGTII does not fall below the base standard of 85.0%.
2. **Maintain Ammonia Synthesis Target:** Operate the synthesis reactors to sustain the **Ammonia Synthesis Target** of 25,000 metric tons per Titan day, ensuring supply covenants with the Mars-orbit agricultural transfer ships are satisfied.
3. **Reclamation Escrow:** Maintain the cash-funded **Environmental Reclamation Reserve** with the Administrative Agent, holding a balance of not less than SC 250,000,000 at all times.

Section 4.02. Negative Covenants. The Borrower shall not, without the explicit written approval of the Administrative Agent:

1. **Geothermal Expansion:** Drill any secondary thermal taps or install supplementary nitrogen-flushing boreholes within a 50-kilometer radius of the primary Dome.

2. **Restricted Payments:** Declare or make any dividend distributions to corporate parent entities if the CGTII is below 87.5% or if a Default Event has occurred and is continuing.

ARTICLE V: DEFAULT EVENTS AND REMEDIES

Section 5.01. Default Events. The occurrence of any of the following events shall constitute an Event of Default:

1. **Tap Blowout or Core Thermal Melt:** Any thermodynamic event causing an uncontrollable thermal blowout at the geothermal borehole, or a total melt of the primary structural water-ice anchors holding the plant chassis.

2. **Covenant Non-Compliance:** Failure to maintain the required balance in the **Environmental Reclamation Reserve** under Section 4.01(3), or failure to meet the **Ammonia Synthesis Target** for more than two consecutive quarters.

3. **Seismic Clathrate Outgassing:** Any seismic shift leading to an uncontained release of gaseous methane/ammonia exceeding 50,000 parts per million in the immediate atmosphere adjacent to the Dome.

XANADU CRYO-CHEMICALS LP

By: _____
Name: Marcus Vance
Title: Managing Director of Subsurface Mining

Date of Execution: **June 26, 2146**

MARTIAN COLONY INVESTMENT SYNDICATE

By: _____
Name: Sarah Vance-Lowell
Title: Senior Investment Partner, Outer-Rim Assets

Date of Execution (the **Signature Date**): **June 26, 2146**